DataML Challenge

Chest X-Ray Multi-Condition Detection

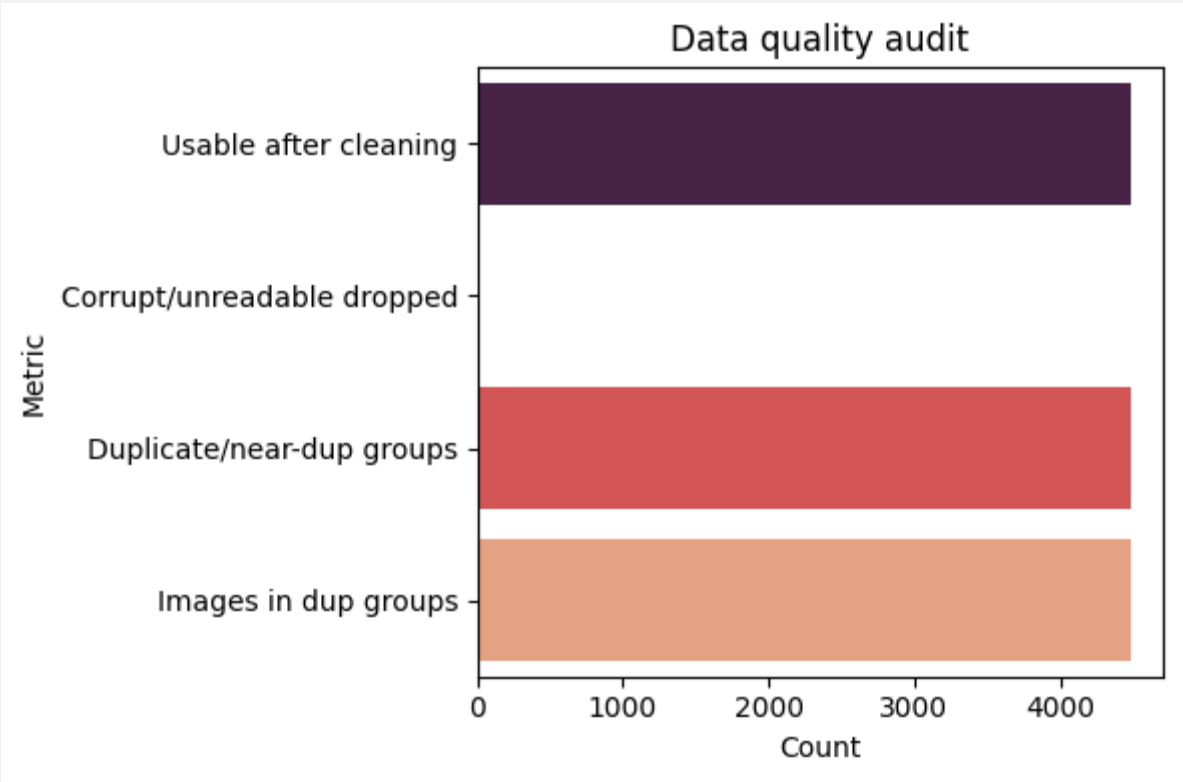
Data Insights & Visualizations

Ashes of the Phoenix

This document walks through what we found while exploring the chest X-ray dataset — data quality, class balance, condition co-occurrence, and a set of deeper visual analyses into what each condition actually looks like in the images. Each section has a placeholder for the corresponding chart and a spot to write down what we observed.

1. Data Quality Check

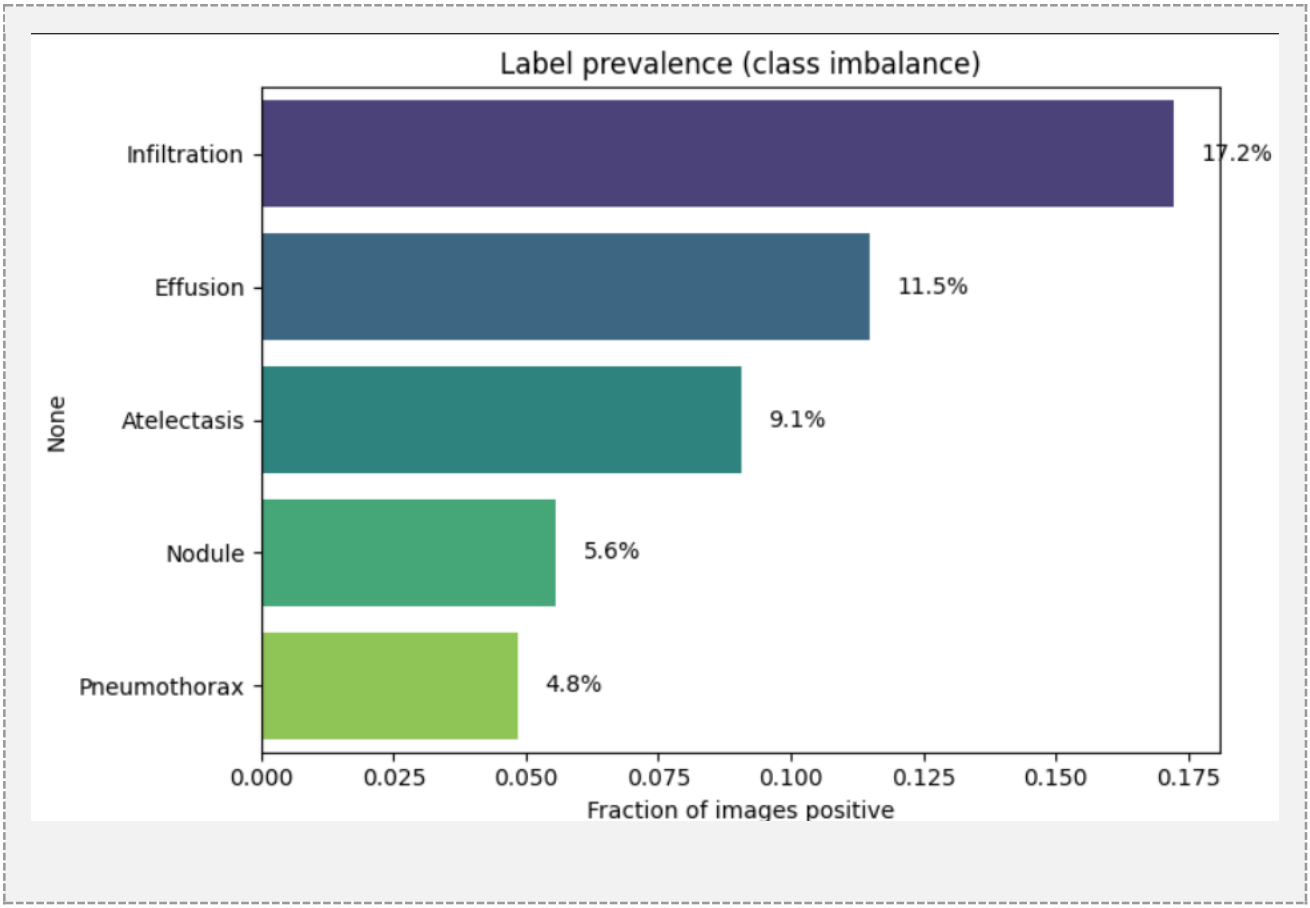
Before looking for patterns, we checked how clean the raw dataset actually was — how many images were corrupt or unreadable, and how many were near-duplicates of each other (which matters because duplicates can quietly leak between training and validation splits).



Our finding: *4,484 of the labeled X-rays were usable after cleaning and none were dropped for being corrupt or unreadable, so this dataset came in clean. The number of duplicate/near-duplicate perceptual-hash groups is essentially equal to the number of usable images, which tells us there was no meaningful image duplication to worry about — so we didn't need to group-constrain our cross-validation folds for leakage from repeated scans.*

2. Class Balance — How Common Is Each Condition?

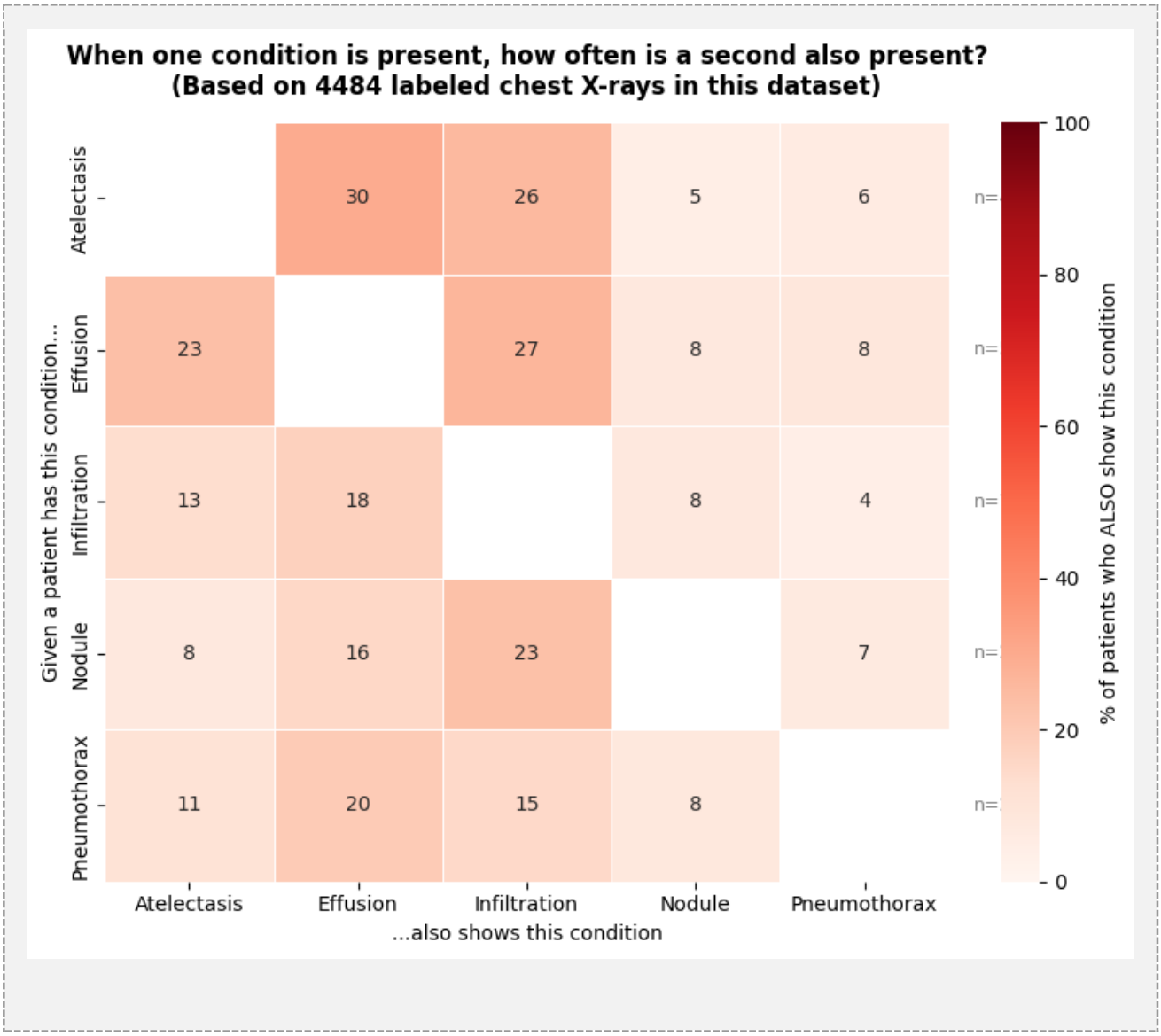
We looked at what fraction of images are positive for each of the five conditions. Chest X-ray datasets are usually dominated by "no finding" cases, so this tells us which conditions we have plenty of examples for and which are rare.



Our finding: *Infiltration is by far the most common finding at 17.2% of scans, followed by Effusion (11.5%) and Atelectasis (9.1%). Nodule (5.6%) and Pneumothorax (4.8%) are much rarer. This is a real class-imbalance problem — our rarest two conditions have roughly a third as many positive examples as Infiltration, which is why we used class-weighted / balanced classifiers and per-condition PR-AUC alongside ROC-AUC rather than relying on accuracy.*

3. Which Conditions Tend to Appear Together?

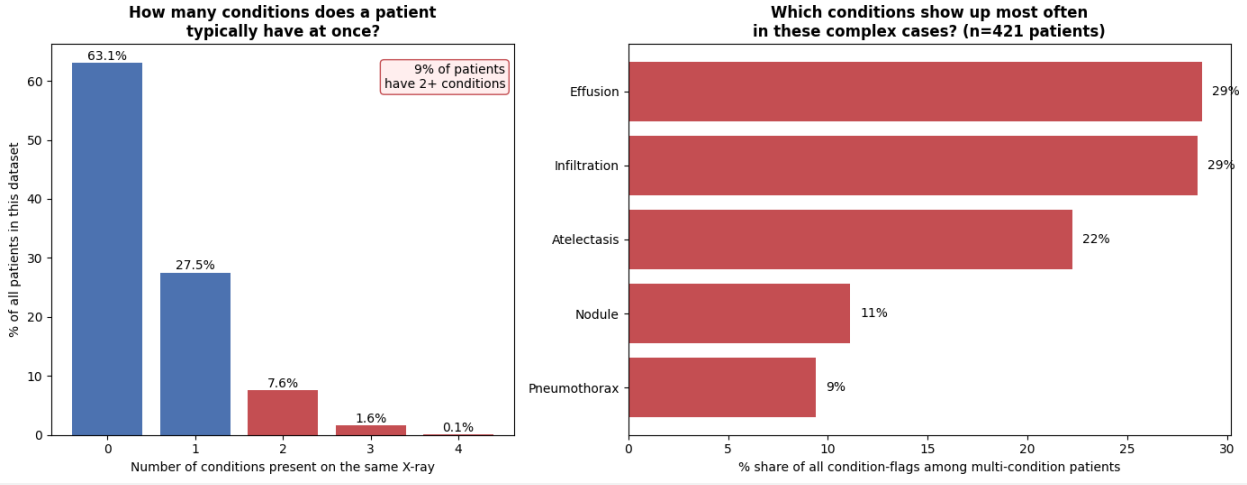
Instead of raw co-occurrence counts (which just reflect how common a condition is overall), we looked at conditional probability: given a patient has condition A, how often do they also have condition B. This is the more meaningful way to see real clinical relationships in the data.



Our finding: *Effusion, Infiltration, and Atelectasis form a tight cluster of comorbidity: for example, patients with Atelectasis also show Effusion 30% of the time and Infiltration 26% of the time, and Effusion/Infiltration co-occur in a similar range. Nodule and Pneumothorax overlap much less with the other conditions (mostly single digits to ~20%), suggesting they tend to present more in isolation. This supports treating the five conditions as correlated but not redundant labels — useful context for why a multi-label-stratified split (Section 8 of the notebook) mattered more than a plain random split.*

4. How Many Conditions Does a Typical Patient Have?

We checked whether most positive X-rays have just one condition flagged, or several at once, and which conditions show up most often in these multi-condition (complex) cases.



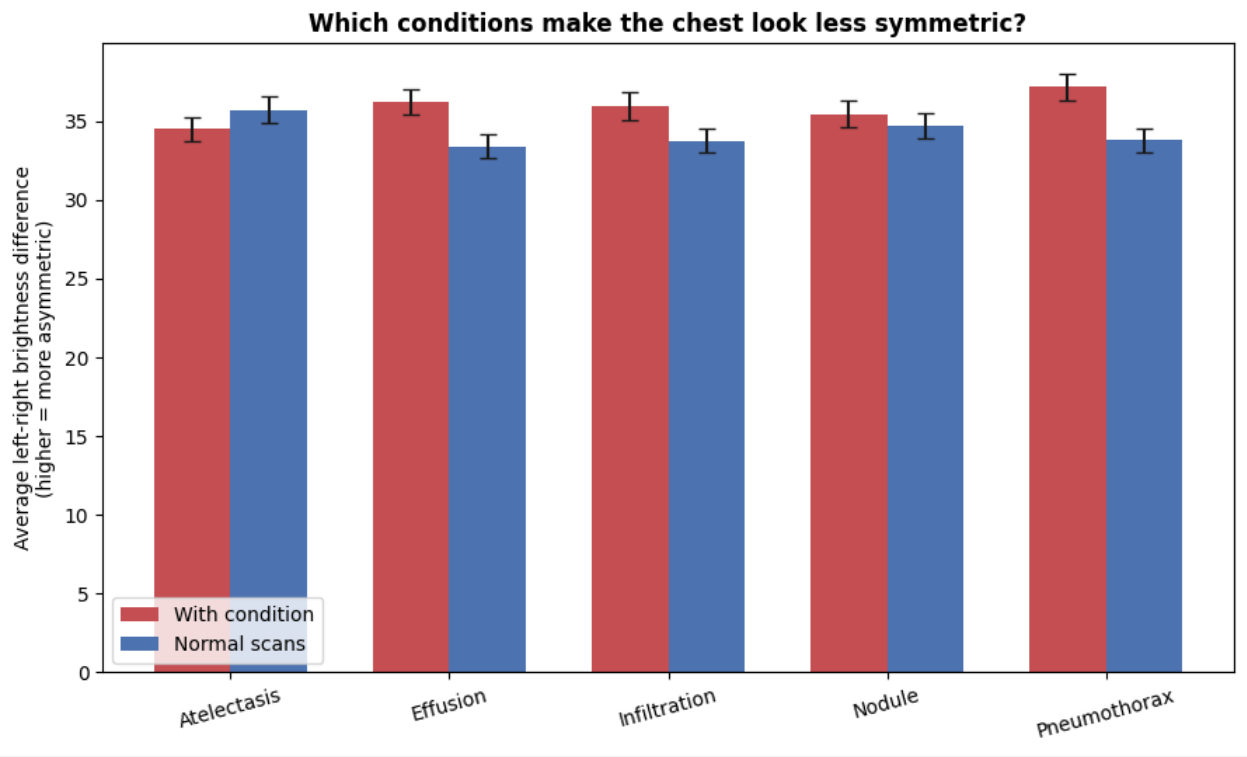
Our finding: *63.1% of scans show no condition and 27.5% show exactly one, so only about 9% of patients have two or more conditions flagged at once — multi-label complexity is real but is a minority of cases. Among those multi-condition patients, Effusion and Infiltration each account for ~29% of all condition-flags and Atelectasis for 22%, meaning these three conditions drive almost all of the comorbid, harder-to-classify cases, while Nodule and Pneumothorax are comparatively rare even within the complex subgroup.*

5. What Do the Conditions Actually Look Like in the Images?

These are deeper visual analyses we ran to understand what each condition changes about the image itself, not just the labels.

5.1 Left-right symmetry

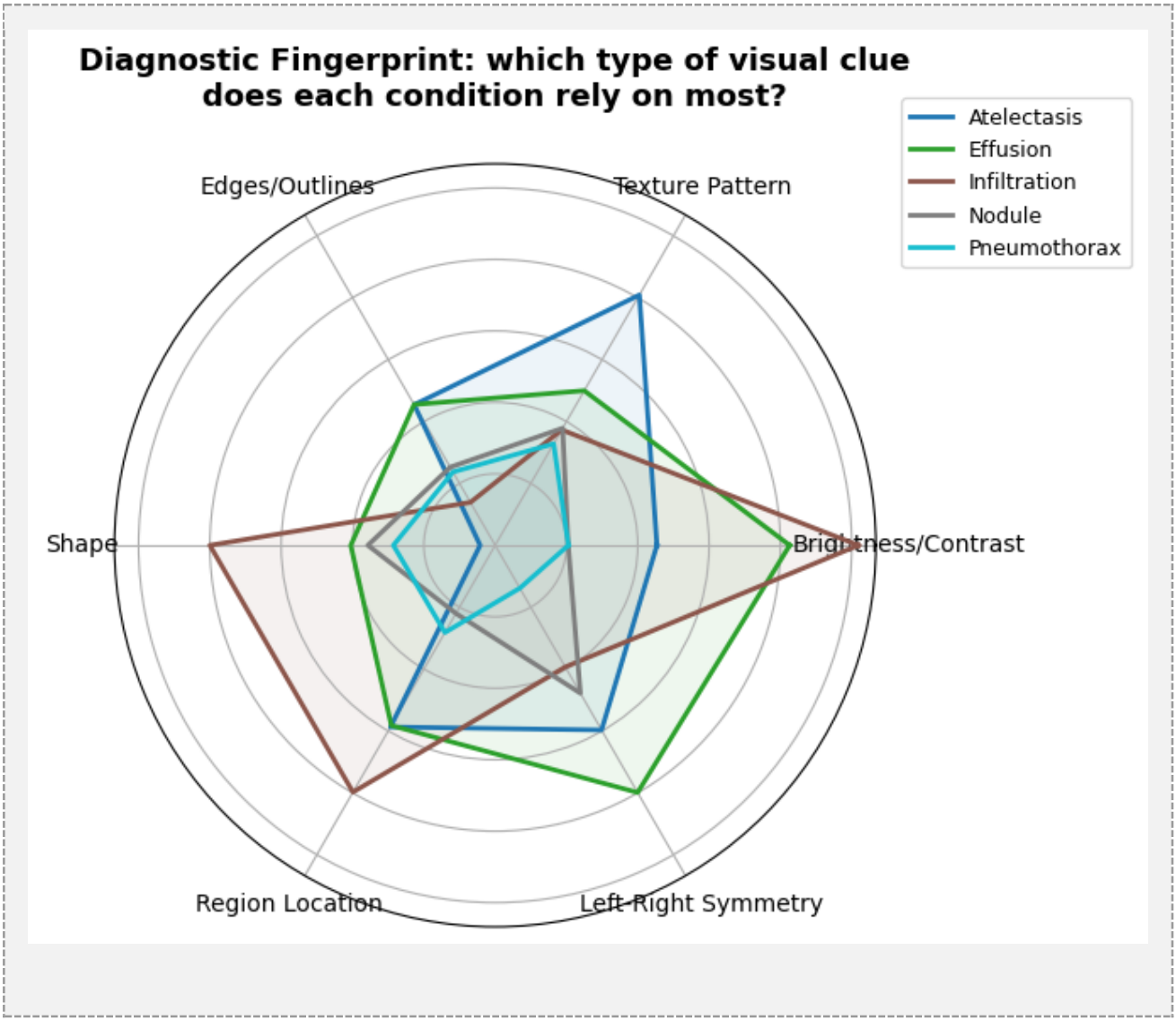
We compared left-vs-right lung brightness for positive vs. normal scans, to see whether a condition tends to show up on one side (asymmetric) or affects both lungs evenly (bilateral).



Our finding: *Pneumothorax shows the biggest gap between positive and normal scans (~37.1 vs ~33.8 average left-right brightness difference), consistent with it typically being a one-sided (unilateral) collapse in real chest X-rays. Effusion and Infiltration also look noticeably more asymmetric than normal scans. Atelectasis is the interesting exception — its asymmetry score is close to, and even slightly below, that of normal scans, suggesting it can present more bilaterally in this dataset, or that a simple brightness-difference measure doesn't fully capture how atelectasis looks.*

5.2 Which visual cue matters most per condition

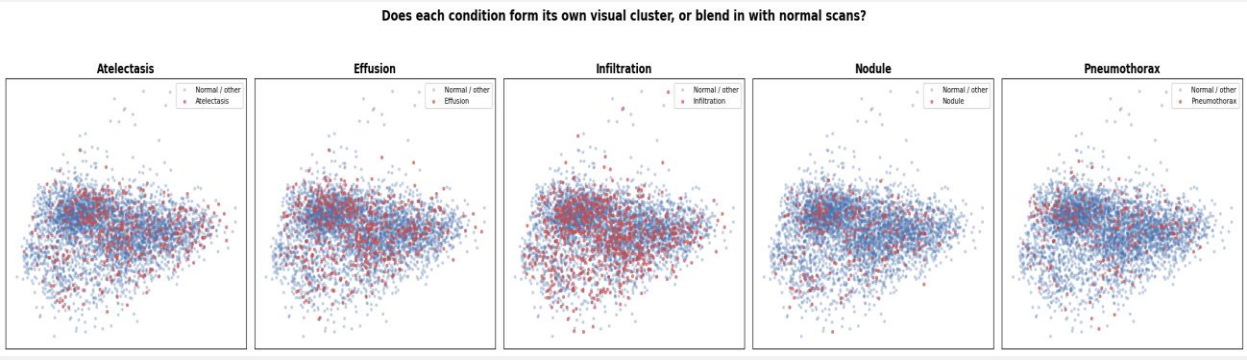
We measured which type of visual feature — texture, edges, brightness, shape, symmetry, etc. — is most useful for telling each condition apart from a normal scan. Different conditions turned out to rely on different types of visual cues.



Our finding: *Different conditions clearly rely on different visual cues: Infiltration stands out on Shape and Region Location (it tends to change where and how densely the lung fields look abnormal), while Atelectasis and Effusion score highest on Brightness/Contrast and Texture Pattern. Nodule and Pneumothorax produce much smaller polygons across every axis, meaning none of our classical hand-crafted feature families captures them strongly — this lines up with them being our hardest conditions to detect in the modeling stage.*

5.3 Do conditions form their own visual cluster?

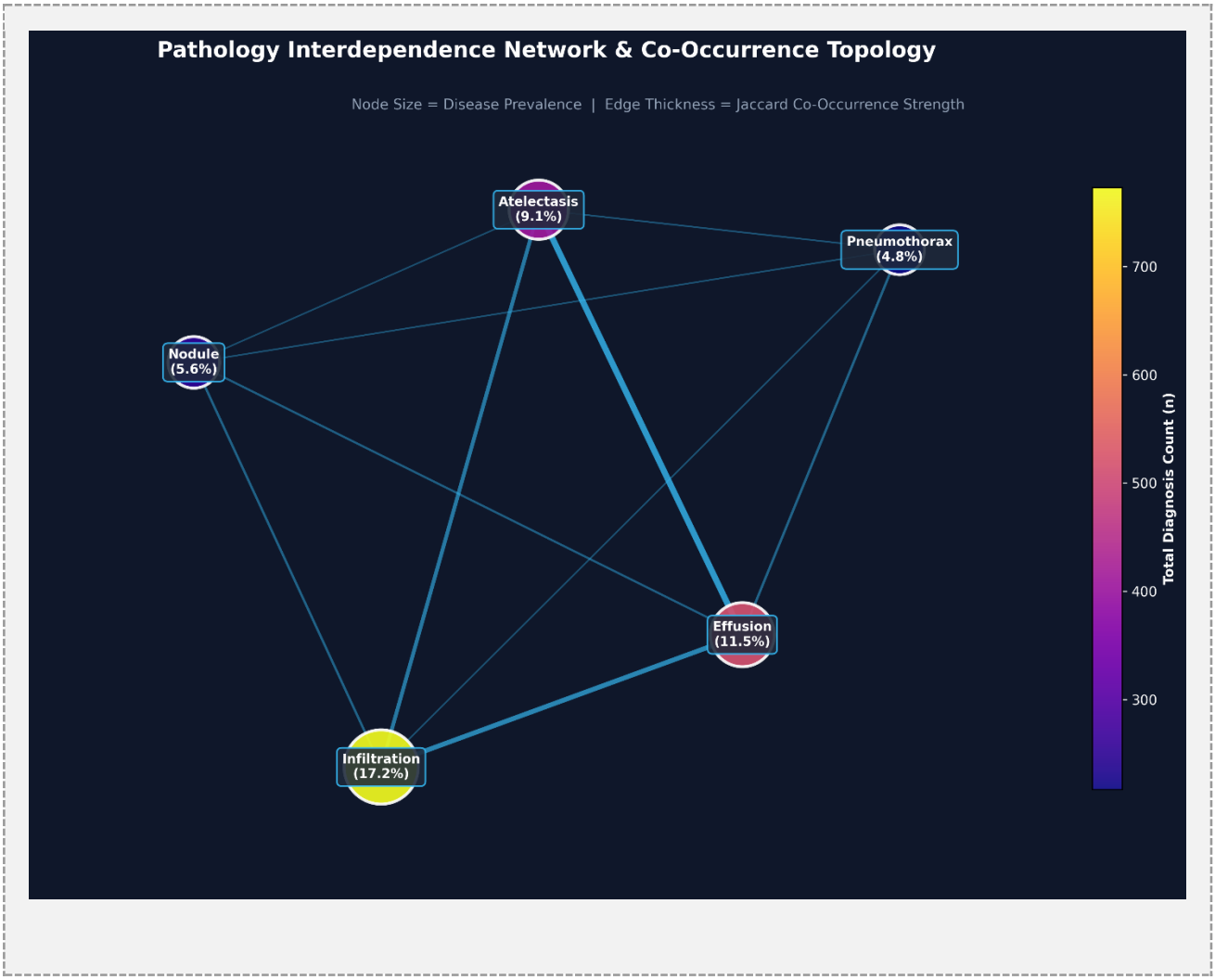
We projected all scans into 2D based on their image features and colored them by condition, to see which conditions look visually distinct from normal scans and which ones blend in and are harder to tell apart.



Our finding: *None of the five conditions forms a clean, separate cluster from normal scans in 2D — the red (positive) points are scattered throughout the blue (normal) point cloud for every condition, confirming that these conditions don't have one dominant, easily separable visual signature. Infiltration and Effusion show the densest concentration of red points near the middle of the cloud, roughly matching their higher AUCs in the modeling stage, while Nodule and Pneumothorax blend in almost completely — reinforcing that they are the hardest conditions for classical features to separate.*

5.4 Condition relationship network

A network graph of the five conditions, where node size reflects how common a condition is and edge thickness reflects how often two conditions occur together — a quick visual summary of which conditions cluster together.



Our finding: *Infiltration is both the most prevalent node and the best-connected one, with thick edges to Atelectasis and Effusion — the Effusion–Infiltration edge is the strongest single relationship in the whole graph. Pneumothorax and Nodule sit at the edges of the network with thinner connections, confirming that they tend to occur more independently of the other four conditions rather than as part of a shared comorbidity pattern.*

6. Summary of What We Found

Pulling together the sections above, in our own words:

Infiltration emerged as the dataset's anchor condition — most prevalent (17.2%), most strongly connected in the co-occurrence network, and tightly clustered with Atelectasis and Effusion, which together drive nearly all of the multi-condition (comorbid) cases. Nodule and Pneumothorax sit at the opposite extreme: rarest by prevalence, weakest in co-occurrence with the other four labels, and hardest to pin down visually, showing the smallest diagnostic-fingerprint footprint and blending almost completely into normal scans in the 2D projection. Pneumothorax was the one clear exception on symmetry — its positive scans showed a distinctly larger left-right brightness gap than normal scans, consistent with its typical presentation as a one-sided lung collapse, whereas Atelectasis showed no unusual asymmetry despite being one of the three most common conditions. Only about 9% of patients had two or more conditions flagged at once, and that complexity was concentrated almost entirely in the Effusion-Infiltration-Atelectasis cluster, confirming these labels are correlated rather than independent and justifying the multi-label-

stratified CV split used in the notebook. Since the raw data itself came in clean (no corrupt images, no meaningful duplication), these patterns reflect genuine class-imbalance and comorbidity structure in the disease labels rather than a data-quality artifact — and they line up with the modeling results, where Infiltration and Effusion were easiest to detect and Nodule and Pneumothorax remained the hardest.